Name : Oviya C M
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DevOps Week 01 Assignment

Introduction:

This document contains the execution of basic Linux commands as part of Week 01 DevOps assignment.

1. Check OS name and version
Command:
hostnamectl

Output:

```
ubuntu@ubuntu:~$ hostnamecrl
Command 'hostnamecrl' not found, did you mean:
  command 'hostnamectl' from deb systemd (255.4-1ubuntu8.12)
Try: sudo apt install <deb name>
ubuntu@ubuntu:~$ hostnamectl
 Static hostname: ubuntu
       Icon name: computer-vm
         Chassis: vm 🖴
      Machine ID: 27b3e2f807a04357aa410736f47aa145
         Boot ID: 0d2d75917d6d48de8e3a44bbc8137a96
  Virtualization: oracle
Operating System: Ubuntu 24.04.4 LTS
          Kernel: Linux 6.17.0-14-generic
    Architecture: x86-64
 Hardware Vendor: innotek GmbH
  Hardware Model: VirtualBox
Firmware Version: VirtualBox
   Firmware Date: Fri 2006-12-01
    Firmware Age: 19y 4month 2w 3d
ubuntu@ubuntu:~$
```

Explanation:

This command displays system information including the operating system name and version.

2. File and Directories

Output:

```
ubuntu@ubuntu:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  snap
ubuntu@ubuntu:~$ mkdir devops_project
ubuntu@ubuntu:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  devops_project  snap
ubuntu@ubuntu:~$ cd devops_project
ubuntu@ubuntu:~/devops_project$ pwd
/home/ubuntu/devops_project
ubuntu@ubuntu:~/devops_project$ cd /home/devops_project
ubuntu@ubuntu:/home/devops_project$ pwd
/home/devops_project
ubuntu@ubuntu:/home/devops_project$ touch plan.txt
touch: cannot touch 'plan.txt': Permission denied
ubuntu@ubuntu:/home/devops_project$ ls -l
total 0
ubuntu@ubuntu:/home/devops_project$ ld -ld
Command 'ld' not found, but can be installed with:
sudo apt install binutils
ubuntu@ubuntu:/home/devops_project$ sudo chown ubuntu /home/devops_project
ubuntu@ubuntu:/home/devops_project$ touch plan.txt
ubuntu@ubuntu:/home/devops_project$ ls
plan.txt
ubuntu@ubuntu:/home/devops_project$ touch network_report.txt
ubuntu@ubuntu:/home/devops_project$ ls
network_report.txt  plan.txt
ubuntu@ubuntu:/home/devops_project$ cp pla.txt /tmp
cp: cannot stat 'pla.txt': No such file or directory
ubuntu@ubuntu:/home/devops_project$ cp plan.txt /tmp
ubuntu@ubuntu:/home/devops_project$ rm devops_project/plan.txt
rm: cannot remove 'devops_project/plan.txt': No such file or directory
ubuntu@ubuntu:/home/devops_project$ rm /devops_project/plan.txt
rm: cannot remove '/devops_project/plan.txt': No such file or directory
ubuntu@ubuntu:/home/devops_project$ ls
network_report.txt  plan.txt
ubuntu@ubuntu:/home/devops_project$
ubuntu@ubuntu:/home/devops_project$ rm -r plan.txt
ubuntu@ubuntu:/home/devops_project$ ls
network_report.txt
ubuntu@ubuntu:/home/devops_project$ cat /tmp
cat: /tmp: Is a directory
ubuntu@ubuntu:/home/devops_project$
```

Explanation:

I created devops_project directory under home, its owned by root so while creating file under the directory I got error as permission denied. Later I changed owner to ubuntu then created files over there. Then, copied plan.txt to /tmp and removed the original file.

3. Users and Permission:

Output:

```
ubuntu@ubuntu:/home/devops_project$ sudo adduser devstudent
info: Adding user `devstudent' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `devstudent' (1002) ...
info: Adding new user `devstudent' (1002) with group `devstudent (1002)' ...
info: Creating home directory `/home/devstudent' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for devstudent
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
info: Adding new user `devstudent' to supplemental / extra groups `users' ...
info: Adding user `devstudent' to group `users' ...
```

```
ubuntu@ubuntu:/home/devops_project$ sudo addgroup devops_team
info: Selecting GID from range 1000 to 59999 ...
info: Adding group `devops_team' (GID 1003) ...
ubuntu@ubuntu:/home/devops_project$ grep devstudent /etc/passwd
devstudent:x:1002:1002:,,,:/home/devstudent:/bin/bash
ubuntu@ubuntu:/home/devops_project$ sudo usermod -aG devops_team devstudent
ubuntu@ubuntu:/home/devops_project$ groups devstudent
devstudent : devstudent users devops_team
ubuntu@ubuntu:/home/devops_project$ touch secrets.txt
ubuntu@ubuntu:/home/devops_project$ ls
network_report.txt  secrets.txt
ubuntu@ubuntu:/home/devops_project$ ls -l
total 0
-rw-rw-r-- 1 ubuntu ubuntu 0 Apr 19 08:08 network_report.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Apr 19 09:14 secrets.txt
ubuntu@ubuntu:/home/devops_project$ sudo chown devstudent secrets.txt
ubuntu@ubuntu:/home/devops_project$ ls -l
total 0
-rw-rw-r-- 1 ubuntu     ubuntu 0 Apr 19 08:08 network_report.txt
-rw-rw-r-- 1 devstudent ubuntu 0 Apr 19 09:14 secrets.txt
ubuntu@ubuntu:/home/devops_project$ chmod 600 secrets.txt
chmod: changing permissions of 'secrets.txt': Operation not permitted
ubuntu@ubuntu:/home/devops_project$ sudo chmod 600 secrets.txt
ubuntu@ubuntu:/home/devops_project$ ls -l
total 0
-rw-rw-r-- 1 ubuntu     ubuntu 0 Apr 19 08:08 network_report.txt
-rw------- 1 devstudent ubuntu 0 Apr 19 09:14 secrets.txt
ubuntu@ubuntu:/home/devops_project$
```

Explanation:

Created user and group, assigned user to group, and restricted file access to owner only.

4. File Compression:

Output:

```
ubuntu@ubuntu:/home/devops_project$ echo "This is log file" > sample.log
ubuntu@ubuntu:/home/devops_project$ ls
network_report.txt  sample.log  secrets.txt
ubuntu@ubuntu:/home/devops_project$ gzip sample.log
ubuntu@ubuntu:/home/devops_project$ ls
network_report.txt  sample.log.gz  secrets.txt
ubuntu@ubuntu:/home/devops_project$ tar -cvf logs.tar sample.log.gz
sample.log.gz
ubuntu@ubuntu:/home/devops_project$
```

Explanation:

Created a log file and compressed it using gzip and archived it using tar command.

5. Package Management:

Output:

```
ubuntu@ubuntu:/home/devops_project$ sudo apt install curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcurl3t64-gnutls libcurl4t64
The following NEW packages will be installed:
  curl
The following packages will be upgraded:
  libcurl3t64-gnutls libcurl4t64
2 upgraded, 1 newly installed, 0 to remove and 196 not upgraded.
Need to get 902 kB of archives.
After this operation, 534 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl4t64 amd64 8.5.0-2ubuntu10.8 [342 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0-2ubuntu10.8 [227 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libcurl3t64-gnutls amd64 8.5.0-2ubuntu10.8 [334 kB]
Fetched 902 kB in 4s (247 kB/s)
(Reading database ... 212669 files and directories currently installed.)
Preparing to unpack .../libcurl4t64_8.5.0-2ubuntu10.8_amd64.deb ...
Unpacking libcurl4t64:amd64 (8.5.0-2ubuntu10.8) over (8.5.0-2ubuntu10.6) ...
Selecting previously unselected package curl.
Preparing to unpack .../curl_8.5.0-2ubuntu10.8_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.8) ...
Preparing to unpack .../libcurl3t64-gnutls_8.5.0-2ubuntu10.8_amd64.deb ...
Unpacking libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.8) over (8.5.0-2ubuntu10.6) ...
Setting up libcurl4t64:amd64 (8.5.0-2ubuntu10.8) ...
Setting up libcurl3t64-gnutls:amd64 (8.5.0-2ubuntu10.8) ...
Setting up curl (8.5.0-2ubuntu10.8) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
```

```
ubuntu@ubuntu:/home/devops_project$ sudo apt install htop
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  lm-sensors
The following NEW packages will be installed:
  htop
0 upgraded, 1 newly installed, 0 to remove and 196 not upgraded.
Need to get 171 kB of archives.
After this operation, 434 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 htop amd64 3.3.0-4build1 [171 kB]
Fetched 171 kB in 2s (89.6 kB/s)
Selecting previously unselected package htop.
(Reading database ... 212676 files and directories currently installed.)
Preparing to unpack .../htop_3.3.0-4build1_amd64.deb ...
Unpacking htop (3.3.0-4build1) ...
Setting up htop (3.3.0-4build1) ...
Processing triggers for desktop-file-utils (0.27-2build1) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

```
ubuntu@ubuntu:/home/devops_project$ sudo apt install net-tools
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  net-tools
0 upgraded, 1 newly installed, 0 to remove and 196 not upgraded.
Need to get 204 kB of archives.
After this operation, 811 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 net-tools amd64 2.10-0.1ubuntu4.4 [204 kB]
Fetched 204 kB in 2s (121 kB/s)
Selecting previously unselected package net-tools.
(Reading database ... 212686 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-0.1ubuntu4.4_amd64.deb ...
Unpacking net-tools (2.10-0.1ubuntu4.4) ...
Setting up net-tools (2.10-0.1ubuntu4.4) ...
Processing triggers for man-db (2.12.0-4build2) ...
```

```
ubuntu@ubuntu:/home/devops_project$ which curl
/usr/bin/curl
ubuntu@ubuntu:/home/devops_project$ which htop
/usr/bin/htop
ubuntu@ubuntu:/home/devops_project$ which net-tools
ubuntu@ubuntu:/home/devops_project$ which ifconfig
/usr/sbin/ifconfig
ubuntu@ubuntu:/home/devops_project$ curl --version
curl 8.5.0 (x86_64-pc-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.13 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.21.2 (+libidn2/2.3.7) libssh/0.10.6/openssl/zlib nghttp2/1.59.0 librtmp/2.3 OpenLDAP/2.6.10
Release-Date: 2023-12-06, security patched: 8.5.0-2ubuntu10.8
Protocols: dict file ftp ftps gopher gophers http https imap imaps ldap ldaps mqtt pop3 pop3s rtmp rtsp scp sftp smb smbs smtp smtps telnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
ubuntu@ubuntu:/home/devops_project$ htop --version
htop 3.3.0
ubuntu@ubuntu:/home/devops_project$ ifconfig --version
net-tools 2.10
ubuntu@ubuntu:/home/devops_project$
```

Explanation:

Installed packages using apt package manager. Used which command to verify installation path. Since net-tools is a package, it was not executed. So, I verified one of its commands ifconfig. Then I used --version to check installed version of software.

6. Networking Commands:

- **ip a**

```
ubuntu@ubuntu:/home/devops_project$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:51:da:1b brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
       valid_lft 70004sec preferred_lft 70004sec
    inet6 fd17:625c:f037:2:773c:df0b:3a49:52df/64 scope global temporary dynamic
       valid_lft 86398sec preferred_lft 14398sec
    inet6 fd17:625c:f037:2:a00:27ff:fe51:da1b/64 scope global dynamic mngtmpaddr
       valid_lft 86398sec preferred_lft 14398sec
    inet6 fe80::a00:27ff:fe51:da1b/64 scope link
       valid_lft forever preferred_lft forever
```

Explanation:

Displays IP address and network interface details.

- **ping**

```
ubuntu@ubuntu:/home/devops_project$ ping google.com -c 4
PING google.com (142.250.193.142) 56(84) bytes of data.
64 bytes from tzdelb-at-in-f14.1e100.net (142.250.193.142): icmp_seq=1 ttl=255 time=99.1 ms
64 bytes from tzdelb-at-in-f14.1e100.net (142.250.193.142): icmp_seq=2 ttl=255 time=194 ms
64 bytes from tzdelb-at-in-f14.1e100.net (142.250.193.142): icmp_seq=3 ttl=255 time=66.3 ms
64 bytes from tzdelb-at-in-f14.1e100.net (142.250.193.142): icmp_seq=4 ttl=255 time=206 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3035ms
rtt min/avg/max/mdev = 66.273/141.185/205.689/59.787 ms
```

Explanation:

Checks connectivity to Google by sending 4 packets.

- **traceroute**

```
ubuntu@ubuntu:/home/devops_project$ sudo apt install traceroute
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  traceroute
0 upgraded, 1 newly installed, 0 to remove and 196 not upgraded.
Need to get 60.5 kB of archives.
After this operation, 162 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 traceroute amd64 1:2.1.5-1 [60.5 kB]
Fetched 60.5 kB in 2s (29.0 kB/s)
Selecting previously unselected package traceroute.
(Reading database ... 212734 files and directories currently installed.)
Preparing to unpack .../traceroute_1%3a2.1.5-1_amd64.deb ...
Unpacking traceroute (1:2.1.5-1) ...
Setting up traceroute (1:2.1.5-1) ...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto mode
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
Processing triggers for man-db (2.12.0-4build2) ...
```

```
ubuntu@ubuntu:/home/devops_project$ traceroute google.com
traceroute to google.com (142.250.193.142), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  0.821 ms  0.659 ms  0.610 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

Explanation:

This traceroute command was used to trace the path to google.com. The first hop shows the gateway, and remaining hops did not respond due to network restrictions.

- **nslookup**

```
ubuntu@ubuntu:/home/devops_project$ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.193.142
Name:   google.com
Address: 2404:6800:4002:809::200e
```

Explanation:

It gives domain name and IP address.

- **netstat**

```
ubuntu@ubuntu:/home/devops_project$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp6       0      0 ::1:631                 :::*                    LISTEN
udp        0      0 0.0.0.0:43982           0.0.0.0:*
udp        0      0 127.0.0.54:53           0.0.0.0:*
udp        0      0 127.0.0.53:53           0.0.0.0:*
udp        0      0 0.0.0.0:5353            0.0.0.0:*
udp6       0      0 :::47255                :::*
udp6       0      0 :::5353                 :::*
ubuntu@ubuntu:/home/devops_project$
```

Explanation:

Displays active network connections and listening ports.